LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 1: Algorithms: Precise Instructions

Module 1 | 45 minutes | Algorithms and sequence | Grades 6–8 · ages 11–14

Objective: Explain what an algorithm is, write a precise sequence of instructions, and debug a sequence that gives the wrong result.

Success criteria

- I can explain what an algorithm is.
- I can write a precise sequence.
- I can debug a sequence by testing one change at a time.

Key vocabulary

- **algorithm**: A precise set of steps to solve a problem.
- **sequence**: The order in which steps run.
- **bug**: A mistake that makes a program behave wrongly.
- **debug**: Find and fix bugs.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Robot teacher (5 min): learners give the teacher instructions to draw a shape on the board; the teacher follows them literally.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Draw with an algorithm: learners follow the steps.
14 min	**Perform**	Code Lab: draw an L, Code Lab: debug the square, How to debug
9 min	**Probe**	Define it, activity
4 min	**Wrap-up**	Exit ticket: Define algorithm in one sentence.

Unplugged warm-up

Robot teacher (5 min): learners give the teacher instructions to draw a shape on the board; the teacher follows them literally.

Teaching the idea

An algorithm is an unambiguous, ordered set of steps. Sequence matters; computers execute literally. Debugging = compare expected and actual behaviour, isolate the first divergence, change one thing, retest. Bridging card links Scratch pen to Python turtle.

Analogy: A GPS route: each turn must be right and in the right order, or you end up somewhere else.

Live demo: Run the buggy square; narrate expected vs actual at each step; fix the third turn live.

Questions to ask

Ask	Listen for
Why do computers need precise instructions?	They do exactly what they're told.
What happens if you swap two steps?	The result can change.
What is the first step of debugging?	Run it and observe.

Differentiation

Support
Pair learners: one reads blocks aloud, one traces on grid paper.

Stretch
Draw your initials using only move and turn blocks.

Common misconceptions

Misconception	How to fix it
The computer knows what you meant	It follows instructions literally.
Debugging means starting again	Change one thing at a time.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Define algorithm in one sentence.

Homework
Write a 6-step algorithm for making a drink. Ask someone to follow it exactly as written.

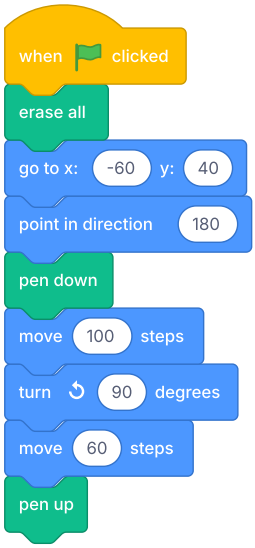
Answer key — Kitabash activities

1. Code Lab: draw an L (perform · code lab)

Goals checked: Draws 2+ lines · Starts clean · Uses turns

One solution:

Pen

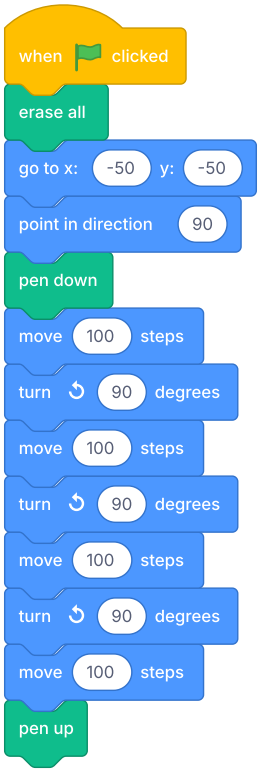


2. Code Lab: debug the square (perform · code lab)

Goals checked: Draws a closed square

One solution:

Pen



3. How to debug (perform · ordering)

Order: Run the program → Compare what happened with what you expected → Find the first step that went wrong → Change one thing → Run it again to test

4. Define it (probe · multiple choice)

Answer: A precise sequence of steps that solves a problem

5. Explain your thinking (probe · free response)

Look for: step, order, precise. **Example:** 1. Pick up the toothbrush. 2. Turn on the tap. 3. Wet the brush. 4. Put toothpaste on the brush. 5. Brush for two minutes. A robot might not know how much toothpaste to use, so step 4 is not precise enough.

Answer key — worksheet

1. **Key words:** algorithm = A precise set of steps to solve a problem. · sequence = The order in which steps run. · bug = A mistake that makes a program behave wrongly. · debug = Find and fix bugs.
2. **Plan your code:** Goals: Draws 2+ lines; Starts clean; Uses turns.

3. **Plan your code:** Goals: Draws a closed square.
4. **How to debug:** 1 Run the program · 2 Compare what happened with what you expected · 3 Find the first step that went wrong · 4 Change one thing · 5 Run it again to test
5. **Define it:** A. A precise sequence of steps that solves a problem — Algorithms exist with or without computers — recipes are algorithms too.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.